Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: COMPILER & DESIGN (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	EduLang: Educational Programming Language Compiler	
S. No.	**University ID**	**Name**
1	2420090024	B.Vishvak Sai
2	2420030207	Akshith
3	2420090086	V.Abhinav
4		
Name of the Guide		Dr. P. Krishna Kishore

ABSTRACT

EduLang: Educational Programming Language Compiler is a lightweight compiler designed to demonstrate and simplify the fundamental concepts of compiler design through a custom educational programming language. The system processes source programs through the major phases of compilation, including lexical analysis, syntax analysis, syntax-directed translation, intermediate code generation, code optimization, and target code generation. The lexical analyzer converts source code into tokens, while the parser verifies grammatical structure and constructs an Abstract Syntax Tree (AST). The AST is translated into Three-Address Code (TAC), which is optimized using techniques such as constant folding, constant propagation, common subexpression elimination, and dead-code elimination. Finally, the optimized intermediate representation is converted into a simplified target instruction set for execution. EduLang provides a practical environment for understanding how high-level programs are transformed into lower-level representations and demonstrates the interaction between different compiler phases.

Signature of the Guide HOD